Emma Ricci-De Lucca

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Machine learning and human-computer interaction researcher with experience in ML development, user research, and digital health innovation. Has led interdisciplinary teams, developed wearable sensing ML models, and designed and evaluated digital health tools.

EDUCATION

Dartmouth College Sep 2023 – Jun 2028 (expected)
PhD Candidate, Human-Computer Interaction & NSF Innovation Fellow, *GPA: 3.9*

Duke University Sep 2021 – Jun 2023
MEng in Biomedical Engineering & Medical Device Design, *GPA: 3.8*

Swarthmore College Sep 2017 – May 2021
BS Engineering & BA French & Francophone Studies, *GPA: 3.7*

SKILLS

Research & UX: Mixed-Methods, Usability Testing, Qualitative Research (Interviews, Surveys, Focus Groups), Thematic Analysis, Participatory Design, Ethnographic Research

AI & ML: scikit-learn, PyTorch, Wearable Sensing & Prediction, Time-Series Modeling, Supervised ML, Data Analysis, TensorFlow

Programming: Python, NumPy, Pandas, HTML, R, D3.js, C/C++

SELECTED ONGOING RESEARCH

Co-Founder, DoulaConnect: AI-Enabled Health Platform Patient Innovation Lab, Geisel School of Medicine

- 1st place among 8 teams and 40 participants at the 2026 Dartmouth Innovation Accelerator for Digital Health pitch competition, earning a $40,000 award judged by top investors and executives. Press coverage here.
- Led an 8-person cross-functional team in designing and evaluating an AI-enabled health platform with 60+ stakeholders (administrators, clinicians, patients). Usability testing showed that participants rated the system near the most usable end of the scale (92% of the possible range), while reporting relatively low workload (30/100).

Machine Learning & Wearable Sensing Empower Lab, Thayer School of Engineering

- Led a 4-person research team developing an ML pipeline that integrates multimodal Fitbit sensor data (heart rate, sleep, activity) with EHR data to investigate digital biomarkers for postpartum depression
- Applied Random Forest classification, SHAP-based interpretability, regression, and mixed-effects modeling to evaluate predictive and statistical relationships between wearable signals and depression

User Research & Human-Centered Design Empower Lab, Thayer School of Engineering

- Led a 2-person research team conducting 14 semi-structured user interviews and inductive thematic analysis in Atlas.ti, synthesizing qualitative insights into design implications for health technologies

PREVIOUS RESEARCH EXPERIENCE

Mechanoacoustic Wearable Project Jan 2025 – Mar 2025

- Built a multimodal wearable combining inertial, mechanoacoustic, and thermal sensors to capture jaw-movement signatures associated with bruxism, with custom firmware and BLE data streaming
- Evaluated sensor placement strategies through multi-channel measurement, identifying configurations for capturing distinct jaw-movement signals

Additive Manufacturing Research, Duke University Jun 2021 – Jan 2022

- Synthesized 181 studies in a systematic review of 3D-printed titanium implants, extracting evidence-based insights to inform design and evaluation of patient-centered healthcare technologies

Computational Microscopy, Swarthmore College Jun 2020 – Aug 2020

- Developed CNN-based computational microscopy models in TensorFlow to reconstruct high-resolution images from low-resolution inputs, integrating deep learning, optics, and optimization

Mechanobiology, University of Pennsylvania May 2019 – Aug 2019

- Applied image analysis and quantitative research methods to microscopy data, contributing to a peer-reviewed publication and presenting findings at BMES 2019

PROFESSIONAL EXPERIENCE

Global Business Development Intern, Siemens Healthineers Feb 2022 – Nov 2022

- Managed Agile portfolios for 9 products in Azure DevOps, modeled 5 business plans, and built 2 Excel dashboards to support SaaS customer acquisition strategy and expansion into 15 new international markets

Digital Health Design Fellow, Duke University Aug 2021 – May 2023

- Selected as 1 of 20 fellows from 91 applicants to conduct ethnographic research with clinicians, translating user insights into prototypes and product strategies to address hospital overcrowding and patient safety

SELECTED PUBLICATIONS

Making Doula Coordination Breakdowns Visible: A Participatory Case Study
Ricci-De Lucca, Chilakapati, Berch, Emerson, Medikondla, Kim, Rastogi, Murnane, & Fortuna
2026, CSCW Workshop Paper (Women's Health as Cooperative Work: Participatory AI Across Care, Community and Policy)

Digital Phenotyping of Perinatal Mental Health: A Systematic Review of Machine Learning and Wearable Sensor Approaches
Ricci-De Lucca, Heavey, & Murnane
2026, Journal of Affective Disorders Reports

Family In The Loop: Enabling Family Involvement in Dementia Care at Long-term Facilities with Person-centered AI Tools
Moore, Tao, **Ricci-De Lucca**, Tadin, Morgan, Nottelmann, & Murnane
2025, CSCW

Wearable Biosensors for Early Detection and Continuous Monitoring of Postpartum Depression: A Human-Centered Approach
Ricci-De Lucca & Murnane
2024, CHI Workshop Paper (PhysioCHI: Towards Best Practices for Integrating Physiological Signals in HCI)

Gaussian Curvature Dilutes The Nuclear Lamina, Favoring Nuclear Rupture, Especially At High Strain Rate
Pfeifer, Tobin, Cho, Vashisth, Dooling, Vazquez, **Ricci-De Lucca**, Simon, & Discher
2022, Nucleus

LEADERSHIP

Graduate Student Mentor, Dartmouth College Sep 2024 – Present

- Mentored 7 undergraduate students, securing a combined $42,500 in competitive research funding across multiple scholarships and fellowships

Thayer School of Engineering Ambassador, Dartmouth College Sep 2024 – Present

- Represented Thayer at recruitment panels, conferences, and student organizations including NSBE, SWE and Tau Beta Pi, engaging 50+ prospective students through 1:1 conversations, campus tours, and admissions events

SELECTED HONORS

Outstanding Graduate Student Mentor, Dartmouth College 2026

1st Place, Dartmouth Innovation Accelerator for Digital Health ($40,000) 2026

Best Rural Healthcare Innovation, New England MedTech Conference 2025

SERVICE

Student Volunteer
SWE 2026 | CHI 2026 | FemTechnology Summit 2023

Evaluator for Peer-Reviewed Journals
CHI 2027 | IMWUT 2026 | CHI 2026 | CHI 2024

Good Beginnings Volunteer, NH 2025 – Present

Animal Welfare Volunteer & Advocate, PA, NC, NH & Italy 2020 – Present